

Programming for Data Science

A comprehensive data collection and exploration
project made in collaboration with:

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Web Scrapping



What we did

Tools + Libraries Used

Selenium: For navigating dynamic pages and handling elements like cookie banners and pagination.

BeautifulSoup: For parsing the HTML and extracting specific content.

Requests: For scraping individual profile pages faster once links were collected.

Workflow

Step
1

Accessed the webpage and handled cookie pop-ups with Selenium

Step
2

Extracted profile links across multiple pages dynamically

Step
3

Scraped professor details using BeautifulSoup and Requests

Step
4

Structured the data into a Pandas DataFrame

Challenges found

And how we dealt with them

Handling dynamic page content and cookie pop-ups

- Used Selenium to interact with the webpage and automate tasks like rejecting cookies and navigating pagination.

Managing duplicates in links and ensuring data integrity

- Used Python sets to remove duplicate links and verified data completeness.

Missing whitespace characters in biography

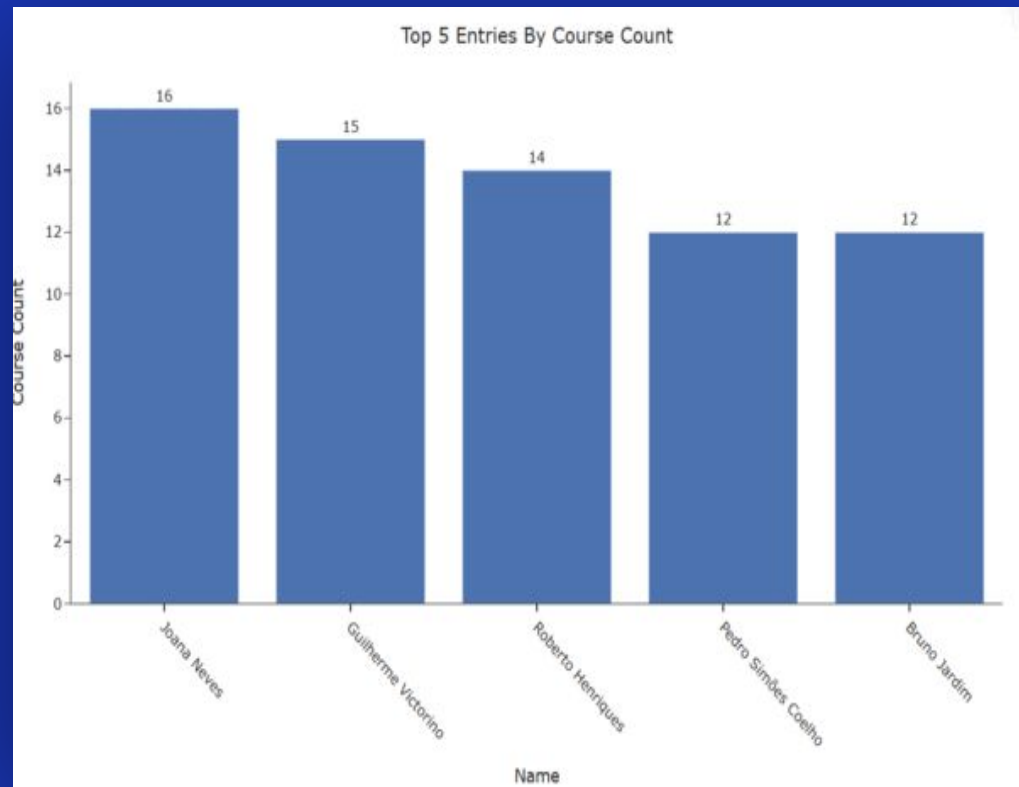
- Switched from Selenium to requests for the final profile page scraping.

Sorting Portuguese names correctly

- Integrated the pyuca library to handle locale-specific sorting.

Data Exploration

Insights gained



Insights Gained

Biographies

Longest Biography: Miguel de Castro Neto (573)

Shortest Biography: Vitor Manita (32)

Courses

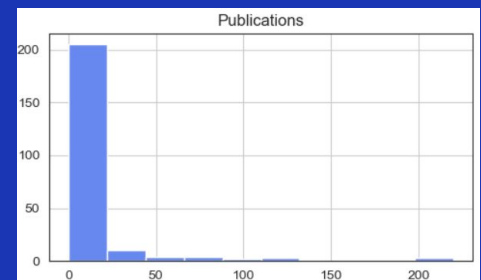
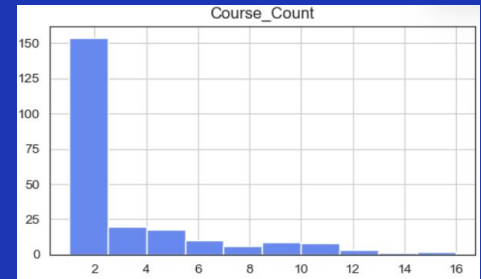
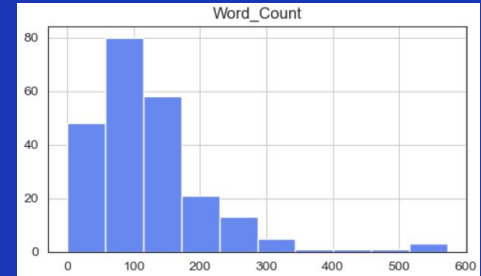
Most Courses Taught: Joana Neves (16)

Least Courses Taught: 110 Different Teachers (1)

Publications

Most Publications: Tiago Oliveira (220)

Least Courses Taught: 137 Different Teachers (0)



Title Distributions

Biographies

Highest AVG Word Count: Full Professor (244)

Lowest AVG Word Count: Research Assistant (61)

Courses

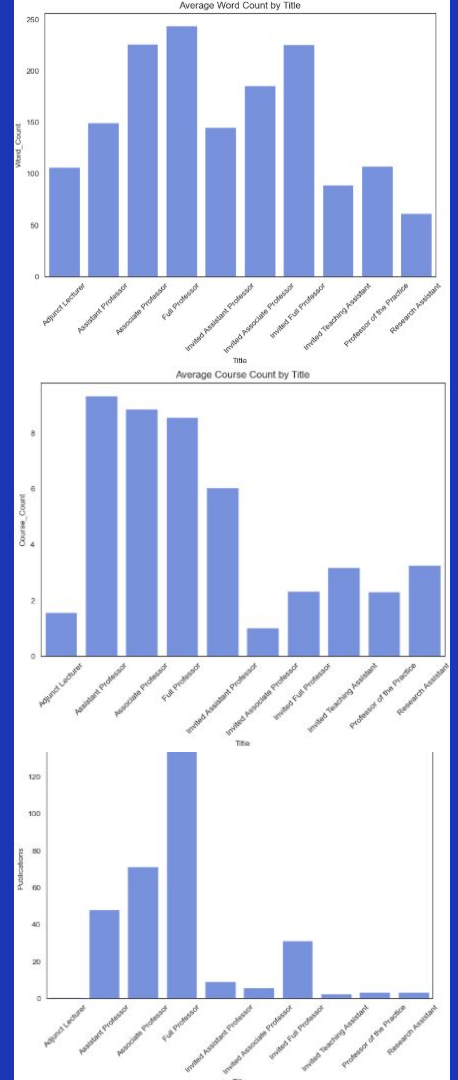
Highest AVG Courses: Assistant Professor (9)

Lowest AVG Courses: Invited Associate Professor(1)

Publications

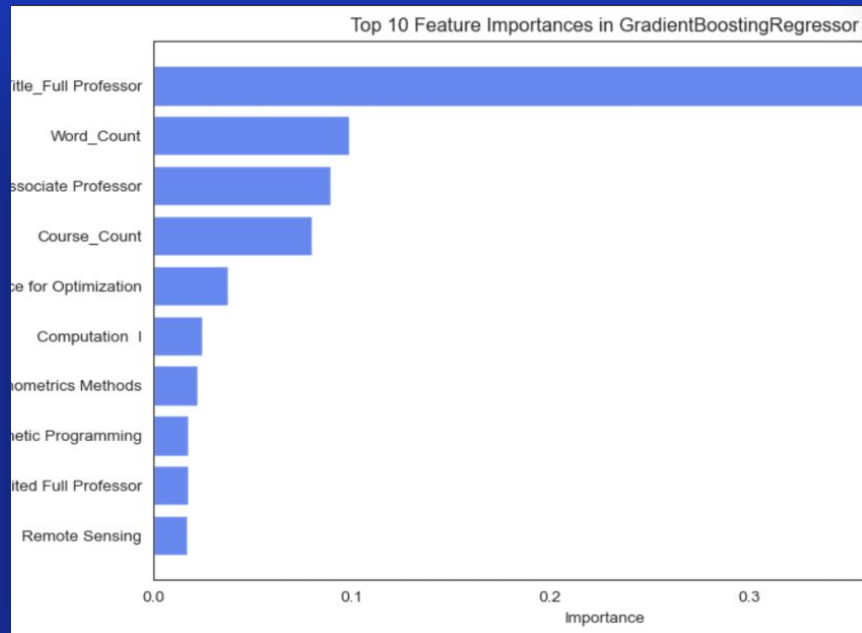
Highest AVG Publications: Full Professor (135)

Lowest AVG Publications: Adjunct Lecturer (1)



Additional Insights

Advanced techniques



Performance Regression

Implementation

Step
1

Feature Selection, Encoding, And Scaling
To maximize our Model functionality and performance

Step
2

Hyperparameter Tuning + Model Initialisation
Using random search and grid search techniques in tandem to maximise performance

Step
3

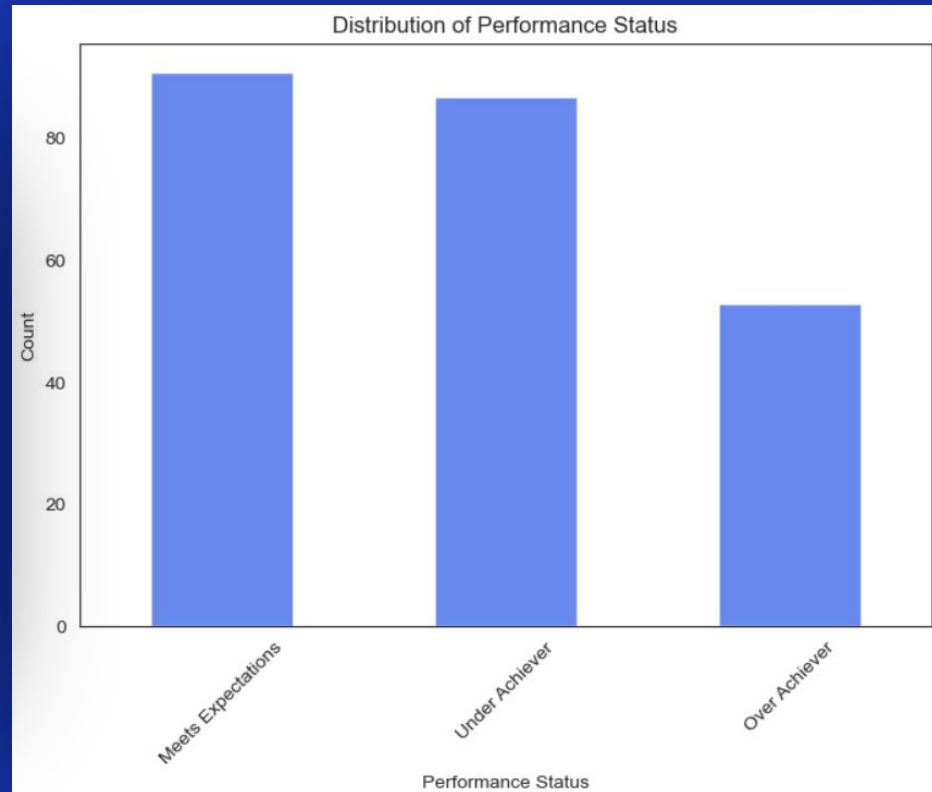
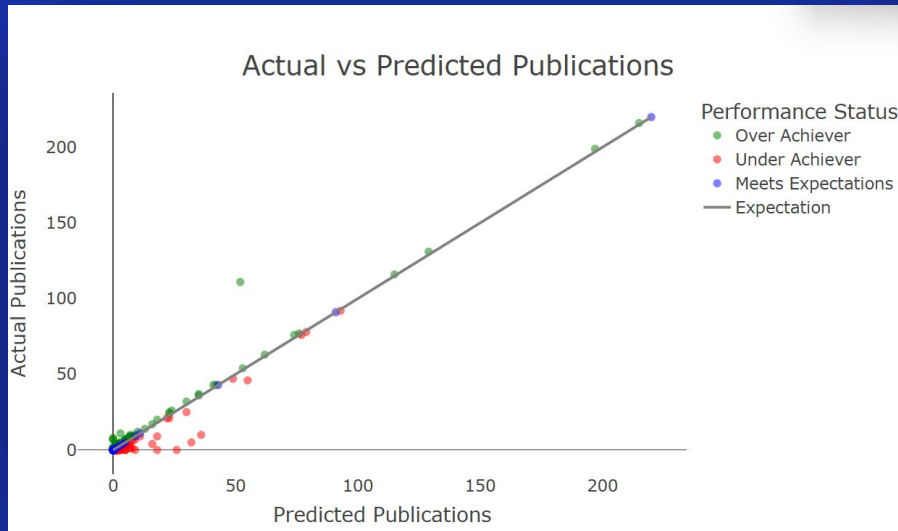
Evaluation And Feature Importances
Getting model performance and feature importance metrics to make changes if needed

Step
4

Classifying And Plotting Results
Using the difference in our predicted values and the actual values to classify performance and plot our findings on a line graph

Performance Regression

Findings



New Course Recommendation

Implementation

Step
1

Course Encoding and Similarity Calculation

Preparing the data by encoding the courses and getting the cosine similarities of the teachers

Step
2

Identify Similar Teachers and Current Courses

For a given teacher, identify other teachers with similar course profiles

Step
3

Generate New Course Recommendations

Generate new courses based on the courses taught by similar teachers

Step
4

Rank and Return Recommendations

Rank the recommended courses by popularity (number of teachers teaching them) and return the most relevant ones.

New Course Recommendation

Sample Usage

```
# Example usage
teacher_id = 5 # Replace with the ID of the teacher for whom you want recommendations
n_recommendations = 3 # Number of desired recommendations
recommended_courses = recommend_courses(teacher_id, n_recommendations)
print(f'Recommended courses for {staff_encoded['Name'][teacher_id]}: {recommended_courses}')
```

```
Recommended courses for Américo Rio: ['Data Management and Storage', 'Armazenamento e Recuperação de Dados', 'UX Research']
```

Biography Word Clouds

Implementation

Step
1

Group Staff By Publications And Course Counts

Split staff into high or low publications and course counts

Step
2

Filter And Preprocess Text Data

Combine all biographies, clean the text, and calculate word frequencies

Step
3

Analyze Top Words

Identify the most common words in the biographies of each group

Step
4

Visualize Findings With Word Clouds

Generate word clouds for each group using their respective word frequencies

